

VISHVESVARAYA TECHNOLOGICAL UNIVERSITY

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Mini Project Report on

“LPG LEAKAGE CONTROL AND ALERT SYSTEM”

Submitted in partial fulfillment of the requirements for the
Second Semester of the Bachelor of Engineering Degree, towards the completion of the
Mini Project under the **Interdisciplinary Project Based Learning Laboratory**,
Department of Basic Sciences.

by

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CERTIFICATE

This is to certify that the File Structures mini project entitled “LPG LEAKAGE CONTROL AND ALERT SYSTEM ” has been successfully carried out by Sahana Ravindra Benageri (1CR25EC181), Vachana Gedela (1CR25CS202), VK Manohar (1CR25CS201 and Sai Sanjay P(1CR25EC182) bonafide students of **CMR Institute of Technology**.

The project is submitted in partial fulfillment of the requirements for the Second Semester of the Bachelor of Engineering Degree, towards the completion of the Mini Project under the **Interdisciplinary Project based learning laboratory, Department of Basic Sciences**.

It is further certified that all corrections and suggestions indicated during the Internal Assessment have been duly incorporated in the project report submitted to the departmental library. This File Structures mini project report has been reviewed and approved as it satisfies the academic requirements prescribed for the said degree.

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1.

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ACKNOWLEDGEMENT

I sincerely express my gratitude to **Dr. Sanjay Jain**, Principal, CMR Institute of Technology, Bangalore, for providing a supportive academic environment.

I extend my thanks to **Dr. Raveesha K H, HoD, Dept. of Physics, CMRIT** for his valuable guidance and support.

I am especially grateful to my internal guide, **Ashutosh Srivastava** Professor, Department of Electronics and Communication and Engineering, for her/his constant encouragement and guidance throughout this project.

I also thank all the faculty members, non-teaching staff, and others who contributed directly or indirectly to the successful completion of this work.

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Abstract

The project titled “LPG Gas Leak Detection and Control System” focuses on enhancing safety in domestic and industrial environments by detecting gas leakage at an early stage and providing immediate alerts. Liquefied Petroleum Gas (LPG) is widely used, but leakage can lead to severe hazards such as fire accidents and explosions.

The system is designed using an MQ-2 gas sensor, Arduino microcontroller, and a GSM module. The gas sensor continuously monitors the concentration of LPG in the surroundings. When the gas level exceeds a predefined threshold, the system triggers a buzzer alarm and sends an alert message to the user via SMS. Additional safety measures, such as activating an exhaust fan or controlling gas supply, can also be integrated.

The methodology involves real-time sensing, data processing using embedded systems, and wireless communication through GSM technology. The system is cost-effective, reliable, and easy to implement.

The outcome of this project is a functional prototype that ensures early detection of gas leaks, minimizes risk, and improves safety by providing timely alerts and preventive actions.

Chapter 1

INTRODUCTION

Gas leakage is one of the major causes of domestic and industrial accidents, especially involving Liquefied Petroleum Gas (LPG). Leakage can lead to fire hazards, explosions, and serious health risks. Traditional safety methods rely on human detection, which is unreliable and often delayed.

This project aims to develop a smart LPG gas leak detection and control system using sensors and microcontrollers. The system detects gas leakage instantly and alerts users through alarms and SMS notifications. It also takes preventive actions like activating exhaust systems.

1.1 Brief History of Gas Detection Systems

Earlier gas detection methods were manual, relying on smell or simple detectors without automation. With the advancement of embedded systems, gas sensors like MQ series enabled real-time detection and monitoring.

1.2 Modern Gas Detection System

Modern systems integrate IoT, GSM communication, and automation, allowing remote alerts and automatic safety actions. This project follows a similar smart approach.

Chapter 2

Problem Statement

2.1 Description

LPG leakage in homes and industries can go unnoticed, leading to dangerous situations. There is a need for a system that can detect leaks early and alert users immediately.

2.2 Challenge Statement

- ☐ Detect gas leakage accurately
- ☐ Provide real-time alerts
- ☐ Work in low-cost environments
- ☐ Ensure fast response time
- ☐ Enable remote notification (SMS/Telegram)

Chapter 3

3.1 Design Thinking Process

a) Empathize:

Users (households, workers) need safety from gas leaks and quick alerts.

b) Define:

Need a reliable, fast, and affordable gas detection system.

c) Ideate:

Ideas included buzzer alarms, mobile alerts, and automatic control systems.

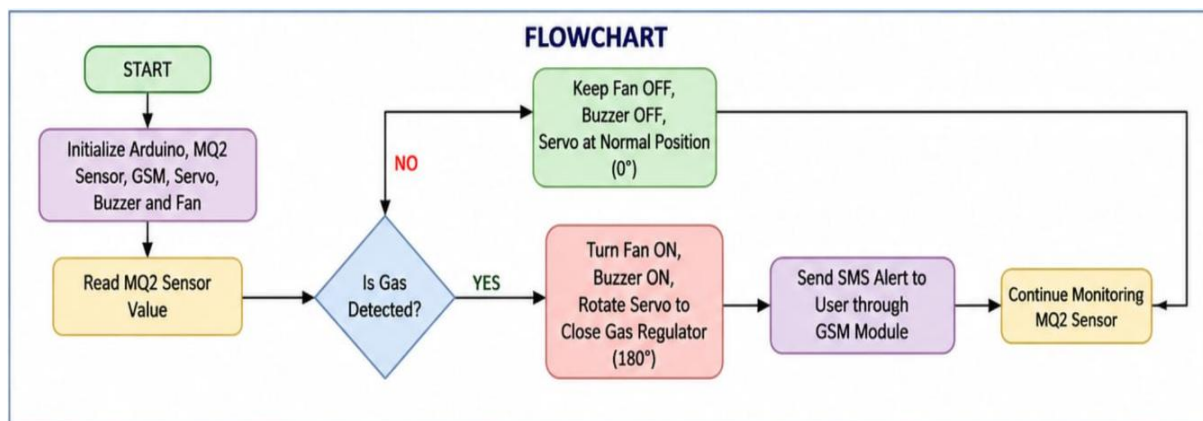
d) Prototype:

Built a system using gas sensor, Arduino, GSM module, and buzzer.

e) Test:

System successfully detected gas and sent alerts within seconds.

3.2 Methodology

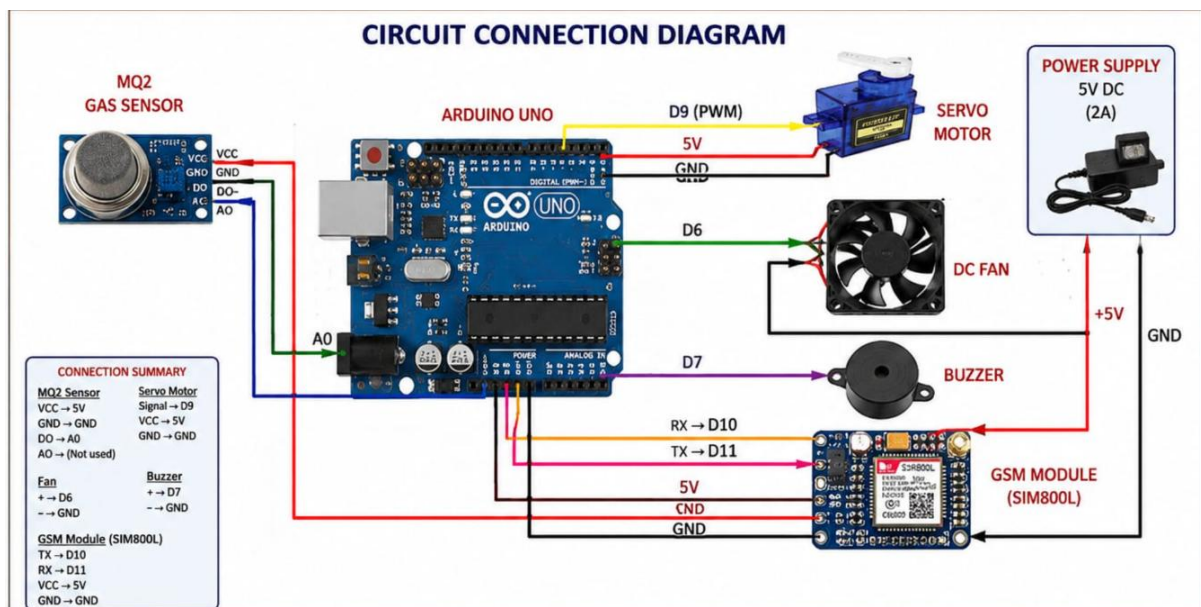


3.3 Prototype Description

3.3.1 Materials Used

- MQ-2 Gas Sensor – Detects LPG gas
- Arduino UNO – Microcontroller
- GSM Module (SIM800/900) – Sends SMS alerts
- Buzzer – Alarm system
- LED – Visual indicator
- Power Supply – 5V

3.3.2 System Diagram



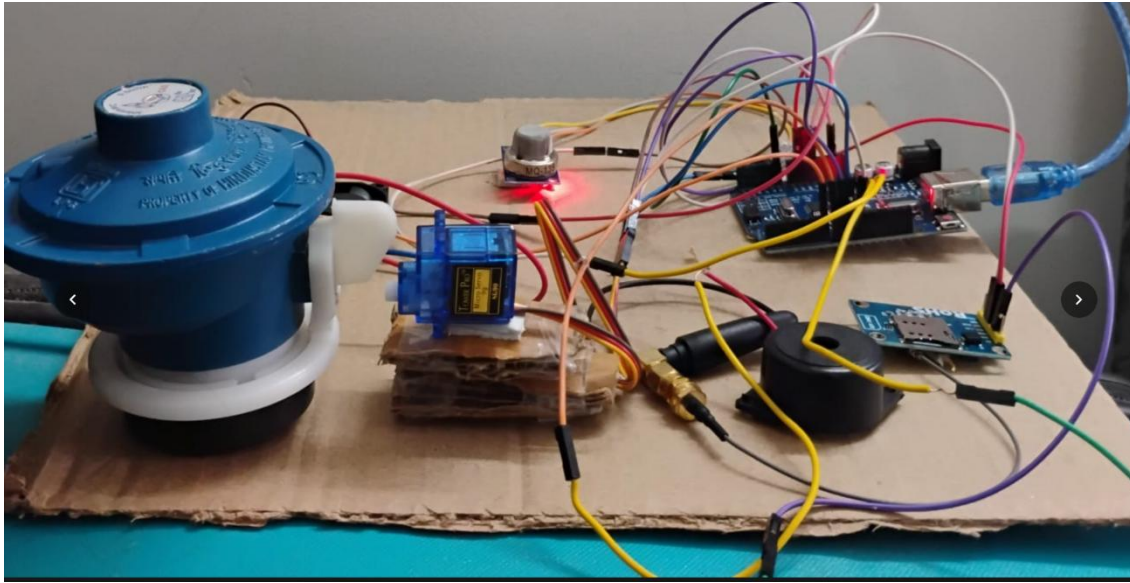
Chapter 4

IMPLEMENTATION

```
1  #include <Servo.h>
2  #include <SoftwareSerial.h>
3
4  Servo myServo;
5
6  // GSM module pins
7  SoftwareSerial gsm(10, 11); // RX, TX
8
9  // Pin definitions
10 int gasSensor = A0;
11 int buzzer = 8;
12 int fan = 7;
13 int threshold = 300;
14
15 bool alertSent = false;
16
17 void setup() {
18
19     pinMode(buzzer, OUTPUT);
20     pinMode(fan, OUTPUT);
21
22     digitalWrite(buzzer, LOW);
23     digitalWrite(fan, LOW);
24
25     myServo.attach(9);
26     myServo.write(0); // Normal position
27
28     Serial.begin(9600);
29     gsm.begin(9600);
30
31     delay(1000);
```

```
59     }
60
61 } else {
62
63     // Normal condition
64     digitalWrite(buzzer, LOW);
65     digitalWrite(fan, LOW);
66
67     // Servo back to normal
68     myServo.write(0);
69
70     alertSent = false;
71 }
72
73 delay(500);
74 }
75
76 void sendSMS() {
77
78     gsm.println("AT+CMGF=1");
79     delay(1000);
80
81     gsm.println("AT+CMGS=\"+919148596954\"");
82     delay(1000);
83
84     gsm.println("Gas Leakage Detected!");
85     delay(1000);
86
87     gsm.write(26);
88
89     delay(3000);
90 }
```

PROTOTYPE MODEL:



Chapter 5

Results and Analysis

User Testing & Feedback

Participants: 5–10 users

Quantitative Results:

- Detection time: ~2 seconds
- SMS alert delay: ~5–10 seconds
- Accuracy: High in closed environment

Qualitative Feedback:

- Users felt safer with alert system
- Easy to use
- Useful for homes and kitchens

Chapter 6

Conclusion & Future Work

The LPG gas leak detection system successfully detects gas leakage and alerts users in real time. It reduces risk and improves safety.

Future Work:

- IoT integration (mobile app)
- Telegram alerts
- Automatic gas valve control
- Cloud monitoring system

References

- ☐ Arduino Documentation
- ☐ MQ-2 Sensor Datasheet
- ☐ GSM Module SIM800 Guide
- ☐ <https://www.arduino.cc>

Annexures

- Circuit diagram
- Code screenshots
- Testing images